THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A horse is tied at a corner of a large square field. The rope is first 14 m long and then lengthened to 42 m. The increase in the area it can graze is: 1

(a) 616 m² **(b)** 1232 m² **(c)** 1386 m² **(d)** 4928 m²

2. A chord of a circle of radius 35 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 753.52 cm² **(b)** 1283.33 cm² **(c)** 1813.15 cm² **(d)** 529.81 cm²

3. A brooch is a circular piece of silver wire of diameter 56 mm. The wire is also used for 8 equal diameters that divide the brooch into 16 sectors. The total length of wire required (in mm) is: 1

(a) 624 mm **(b)** 400 mm **(c)** 2464 mm **(d)** 176 mm

4. A horse is tied at the midpoint of one side of a large square field with a rope 56 m long. The area of the field it can graze is: 1

(a) 176 m² **(b)** 9856 m² **(c)** 88 m² **(d)** 4928 m²

5. A sector of a circle of radius 42 cm has central angle 105°. The perimeter of the sector is: 1

(a) 119 cm **(b)** 161 cm **(c)** 203 cm **(d)** 77 cm

6. An arc of length 88 cm subtends an angle at the centre of a circle of radius 42 cm. The angle subtended by the arc at the centre is: 1

(a) 60° **(b)** 120° **(c)** 150° **(d)** 240°

7. A square is inscribed in a circle of radius 56 cm. The area of the circle outside the square is: 1

(a) 6720 cm² **(b)** 6272 cm² **(c)** 9856 cm² **(d)** 3584 cm²

8. A horse is tied at a corner of a large square field. The rope is first 28 m long and then lengthened to 42 m. The increase in the area it can graze is: 1

(a) 3080 m² **(b)** 1386 m² **(c)** 770 m² **(d)** 154 m²

9. **Assertion (A):** The area of a ring between two concentric circles of radii 14 cm and 7 cm is $462\ \text{cm}^2$ (use $\pi = \frac{22}{7}$). 1

Reason (R): The area of a ring between concentric circles of radii R and r $(R > r)$ is $\pi(R^2 - r^2)$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** In 30 minutes the minute hand, 7 cm long, of a clock sweeps an area of $77\ \text{cm}^2$ (use $\pi = \frac{22}{7}$). 1

Reason (R): The minute hand of a clock turns through 90° in 30 minutes.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A circular park of radius 60 m is surrounded by a path 7 m wide. Find the area of the path. 2

12. An umbrella has 24 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 84 cm. Find the area between two consecutive ribs. 2

13. A wheel has radius 42 cm. Find the number of complete revolutions it makes to cover 198 m. 2

14. A circular park of radius 28 m is surrounded by a path 3.5 m wide. Find the area of the path. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A sector of a circle of radius 42 cm has central angle 170°. Find the area of the sector. 3

16. A quadrant of a circle of radius 91 cm is cut from a square of side 91 cm, with the centre of the quadrant at one corner of the square. Find the area of the remaining part of the square. 3

17. The minute hand of a clock is 42 cm long. Find the area swept by the minute hand in 35 minutes. 3

OR

A sector of a circle of radius 42 cm has central angle 200°. Find the area of the sector. 3

18. A quadrant of a circle of radius 98 cm is cut from a square of side 98 cm, with the centre of the quadrant at one corner of the square. Find the area of the remaining part of the square. 3

OR

The minute hand of a clock is 42 cm long. Find the area swept by the minute hand in 40 minutes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A circular running track has an inner boundary of radius 28 m and an outer boundary of radius 35 m, so the track is 7 m wide. (Take $\pi = \frac{22}{7}$.) 5

(i) Find the area enclosed by the inner circle.

(ii) Find the area enclosed by the outer circle.

(iii) Find the area of the track.

(iv) Find the cost of paving the track at ₹ 25 per square metre and the length of the inner boundary.

20. A chord of a circle of radius 70 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

OR

Two concentric circles have radii 21 cm and 14 cm. A sector of angle 180° is marked in both. Find the area of the region between the two arcs within the sector. 5

END OF QUESTION PAPER — **Total: 40 Marks**